Flipkart Sales Analysis using PySpark in Databricks

Introductions PySpark:

- PySpark is a Python API for Apache Spark. It is used to process, clean, transform, and analyze large datasets efficiently.
- Databricks is a cloud-based platform that supports Apache Spark and helps us perform data analysis using PySpark.
- In this project, we use PySpark to analyze a Flipkart sales dataset containing 200 rows and 10 columns.

Objectives

- Load the Flipkart sales dataset.
- Understand the dataset structure.
- Check and handle missing values.
- Remove duplicate records.
- Perform data cleaning and transformation.
- Analyze sales by category, product, and city.
- Identify top-selling products.
- Create visualizations to understand sales performance.

Tools Used

- Platform: Databricks
- Programming Language: Python
- Data Processing: PySpark
- Dataset: Flipkart Sales Dataset (200 rows, 10 columns)

1. Databricks notebook structure

Create these five notebooks in the following order.

01_Data_Ingestion

Load the CSV file and inspect the data.

02_Data_Cleaning

Handle missing values, duplicates and data types.

03_Data_Transformation

Calculate sales, discounts, and time features.

04_Business_Analysis

Calculate KPIs and analyze sales performance.

05_Visualization_Dashboard

Create charts for your final dashboard.

2. Dataset columns

Column	Purpose
Order_ID	Unique order identifier
Customer_ID	Customer identifier
Product_Category	Product category
Product_Name	Product name
Order_Date	Date of order
Quantity	Units purchased
Unit_Price	Price per unit
Discount_Percent	Discount percentage
City	Customer/order city
Payment_Method	Payment type

3. Notebook 01 — Data Ingestion

Objective: Load the CSV file into a PySpark DataFrame and inspect the data.

First, upload your CSV to Databricks using Catalog / Add data / Upload files to a volume or the upload option available in your workspace. Update file_path to the actual location of your uploaded file.

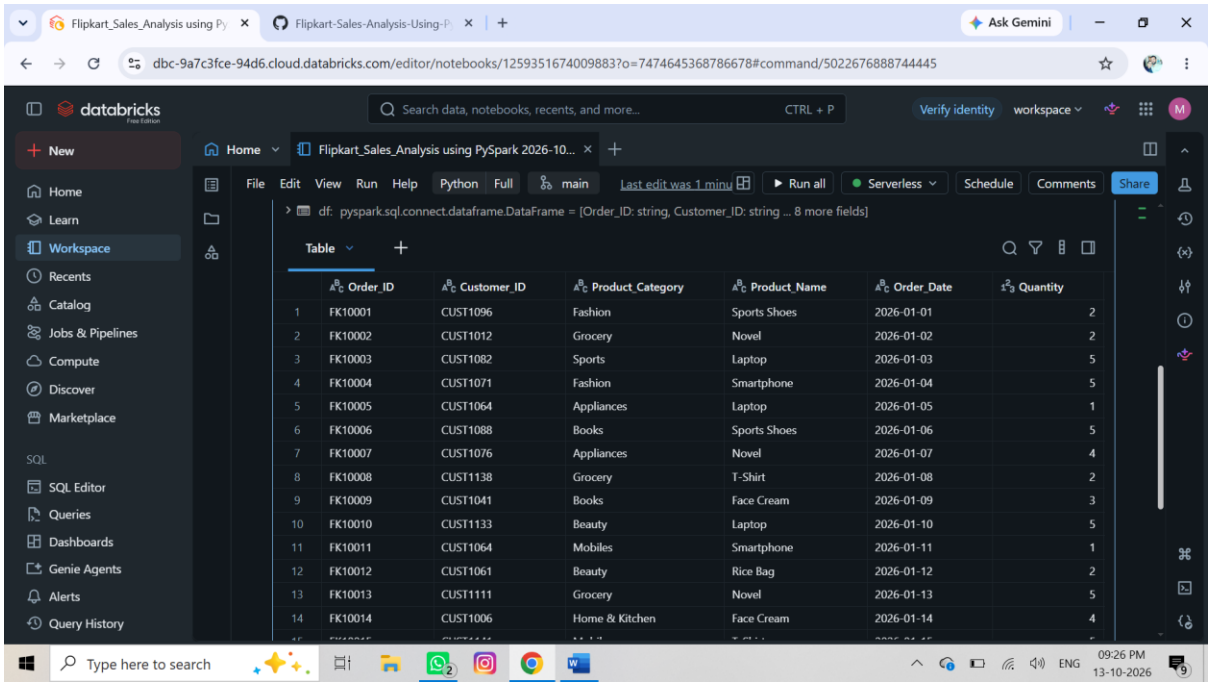
Cell 1 — Import PySpark functions

```
rom pyspark.sql import functions as F
from pyspark.sql.types import *
```

Cell 2 — Load the dataset

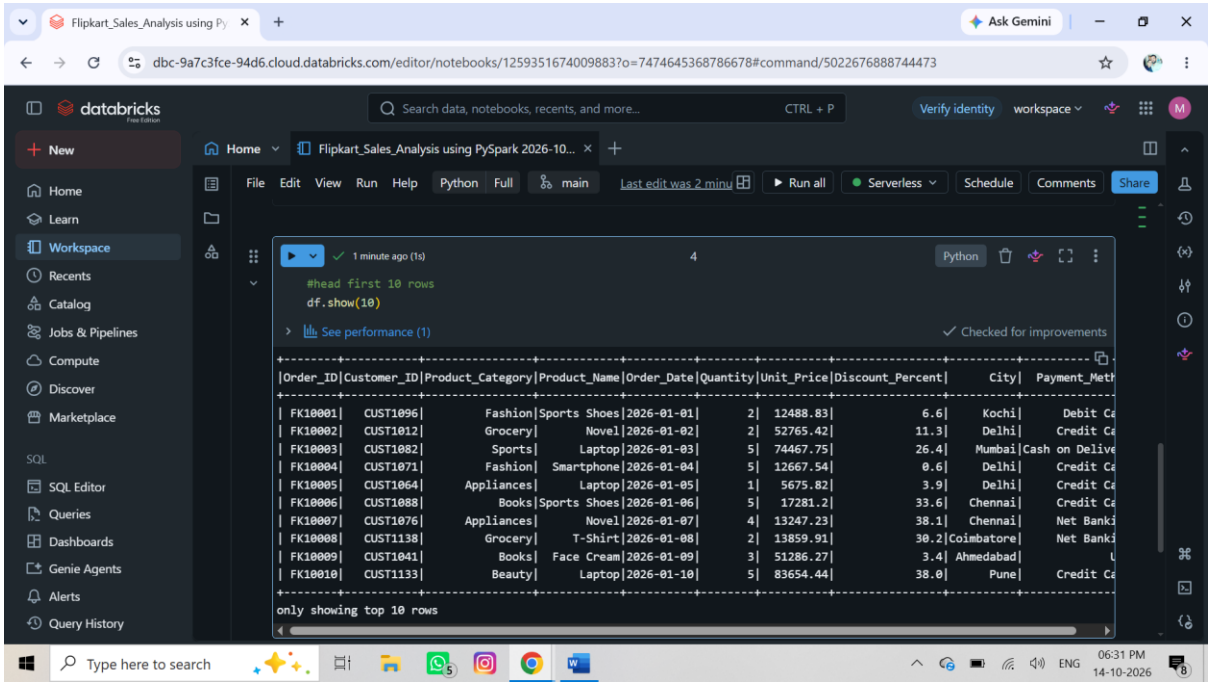
```
from pyspark.sql import functions as F
import pandas as pd

file_path = "https://github.com/mskalaiwani00294-lang/Flipkart-Sales-Analysis-Using-PySpark-Databricks/raw/refs/heads/main/flipkart_style_dataset_200_rows.csv"
df = spark.createDataFrame(pd.read_csv(file_path))
display(df)
```



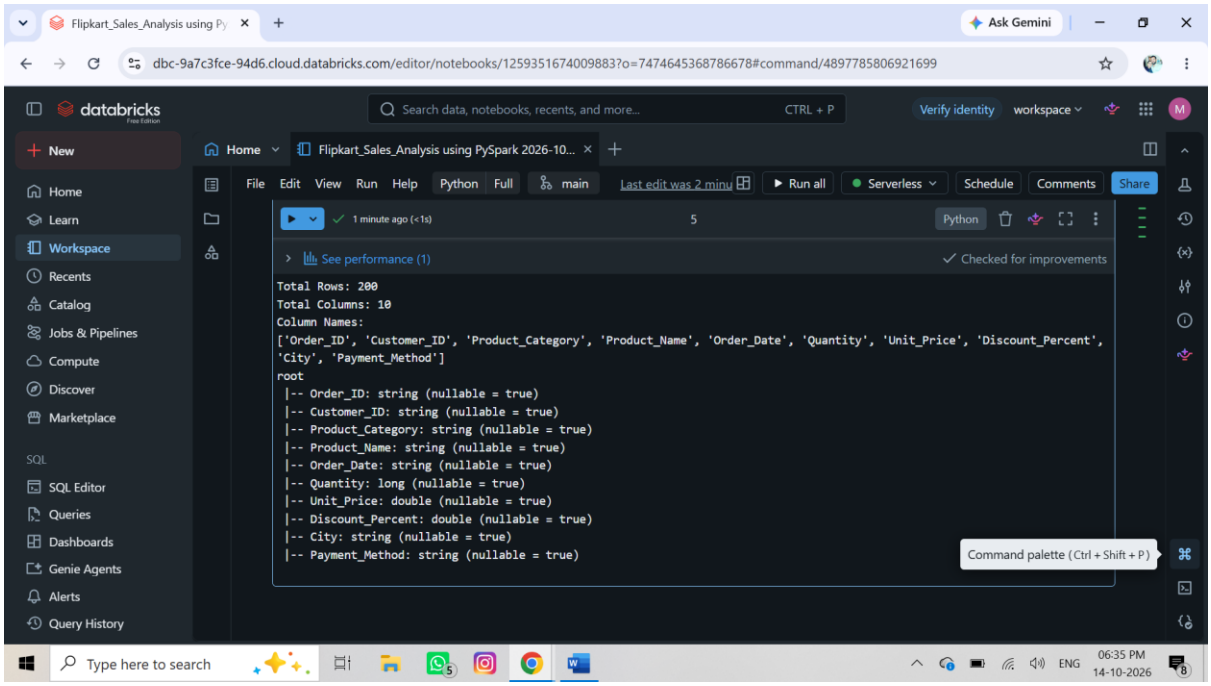
Check Head

df.show(10)



Cell 3 — Check rows, columns and schema

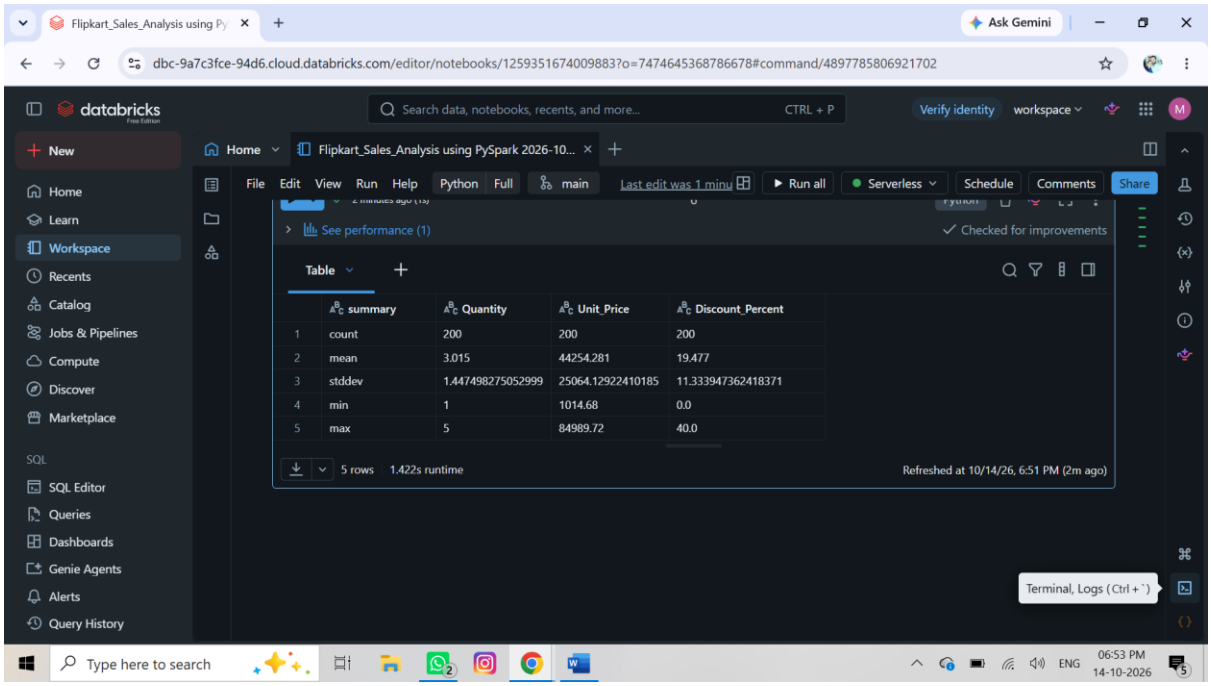
```
print("Total Rows:", df.count())
print("Total Columns:", len(df.columns))
print("Column Names:")
print(df.columns)
df.printSchema()
```



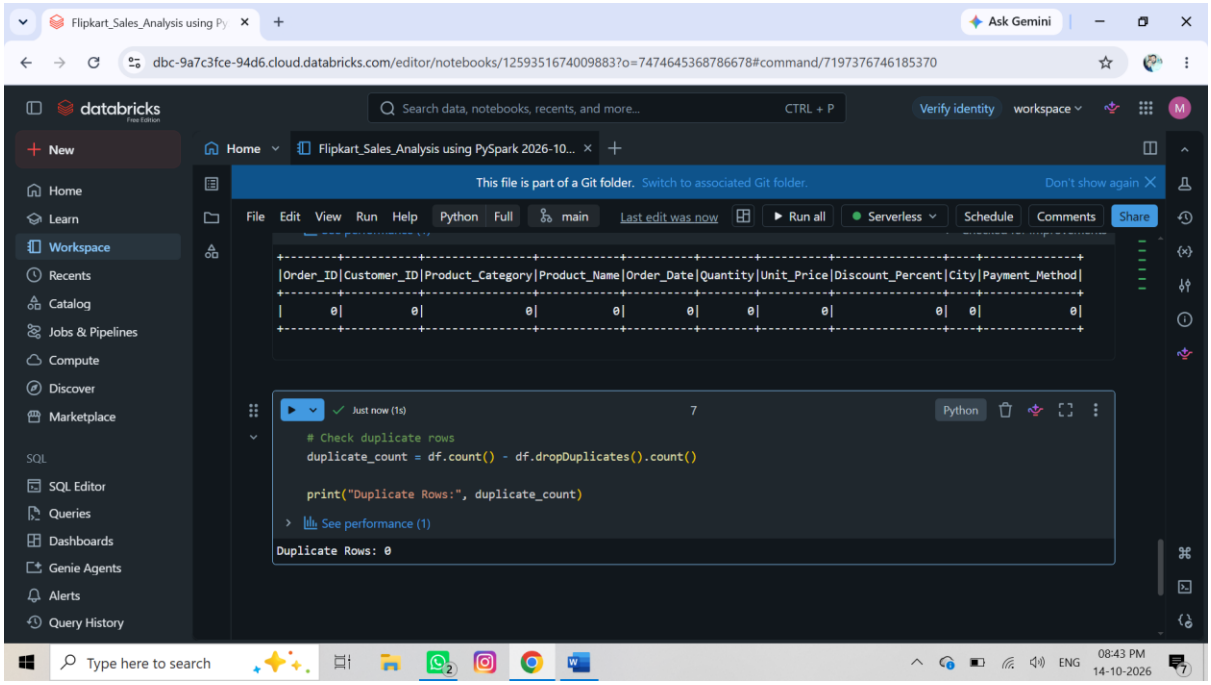

```python
from pyspark.sql.types import NumericType

# Select only numerical columns
numeric_cols = [
    field.name
    for field in df.schema.fields
    if isinstance(field.dataType, NumericType)
]

# Display descriptive statistics
df.select(*numeric_cols).describe().display()
```



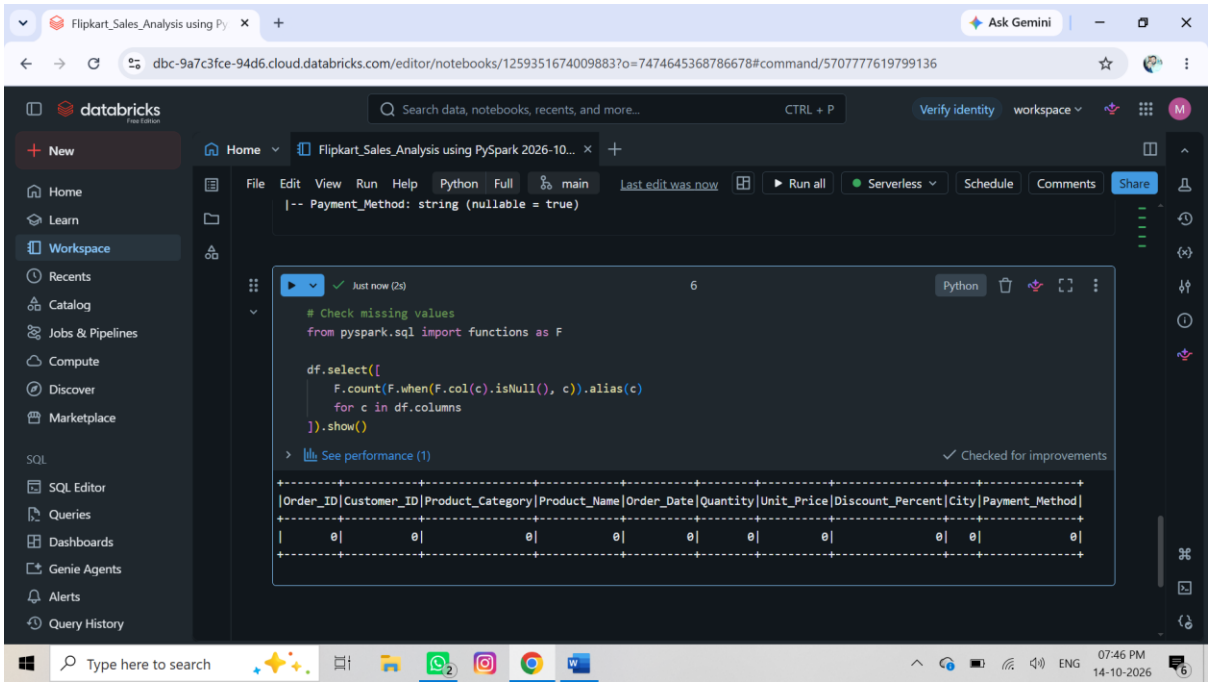
Check Duplicated value



4. Notebook 02 — Data Cleaning

Objective: Remove duplicate orders, handle missing values and correct data types.

Cell 2 — Check missing values



F.col(c)-Selects a column

.isNull()-Checks for NULL values

F.when()-Returns 1 when the condition is true

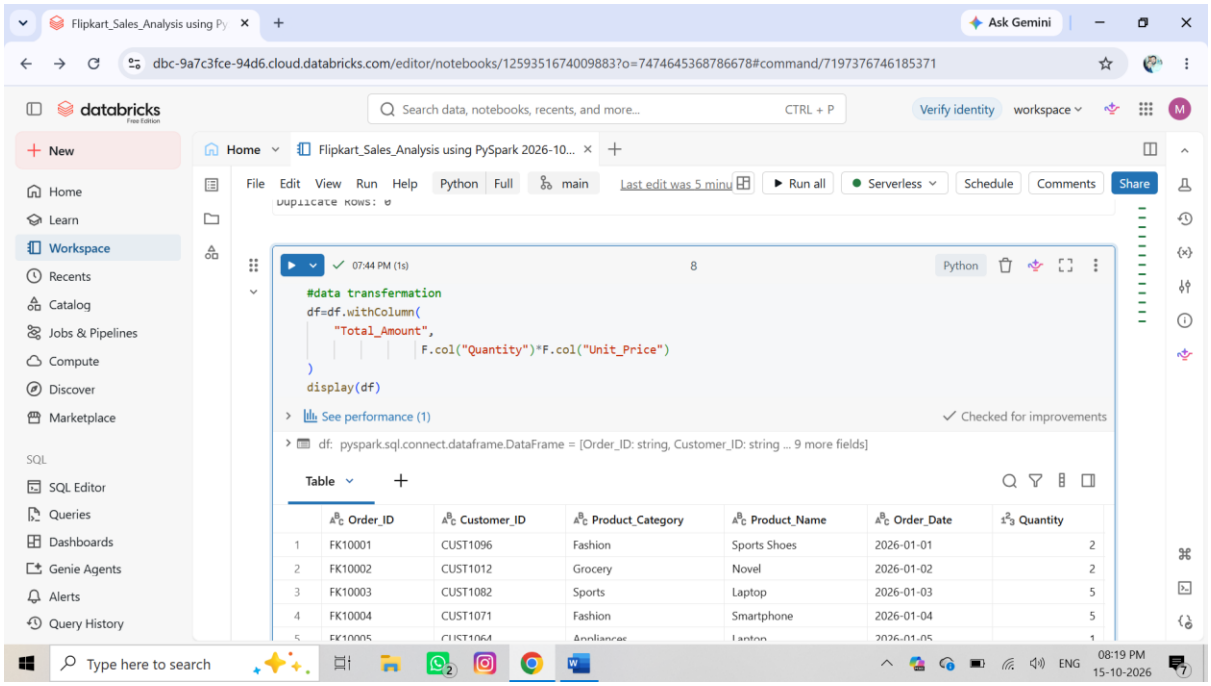
F.count()-Counts non-NULL results

.alias(c)-Names the output column

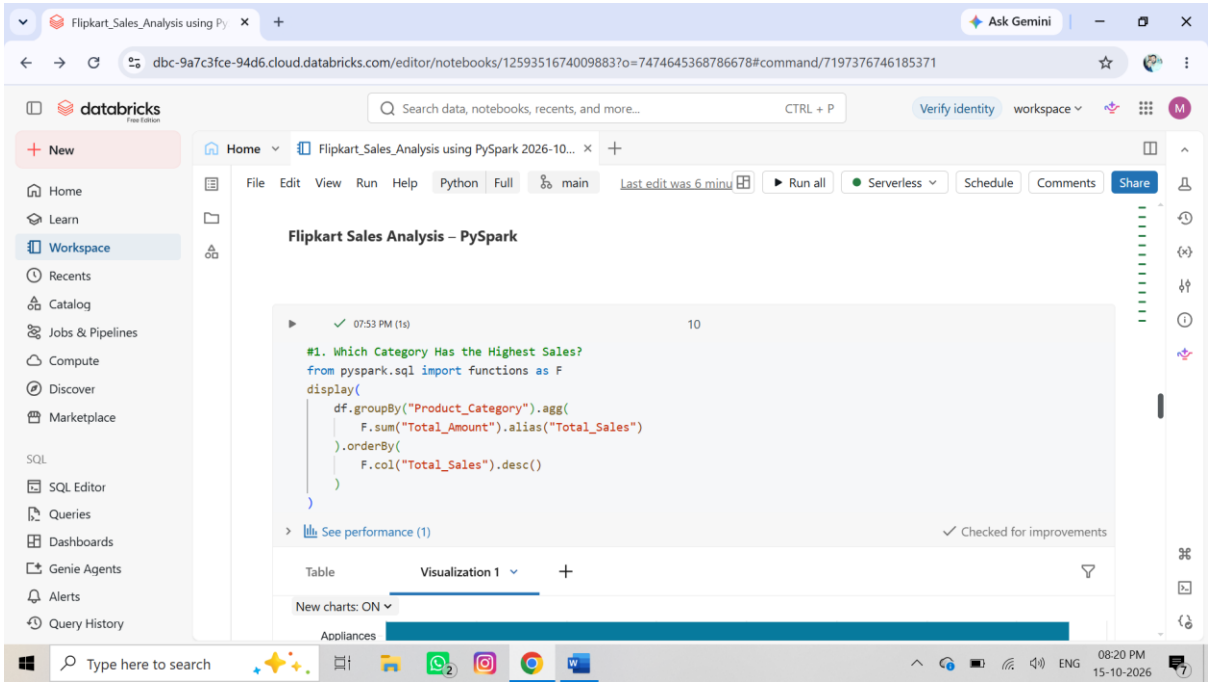
df.columns-Returns all column names

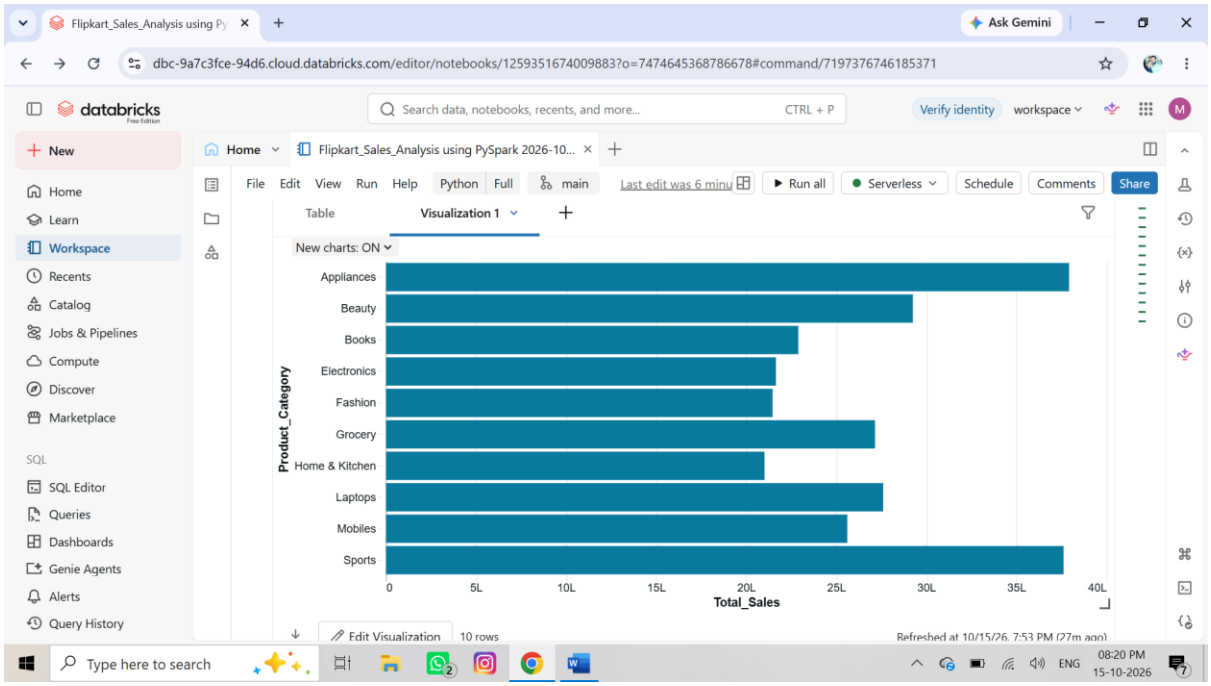
.show()-Displays the result

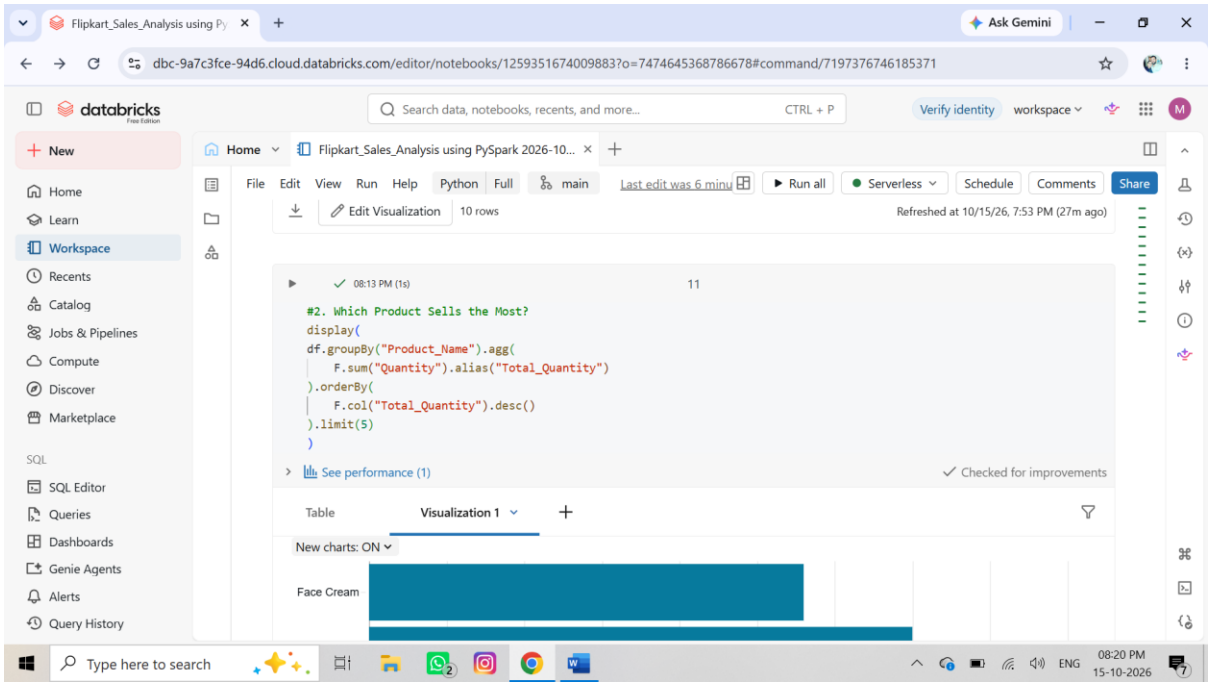
Data Transformation:

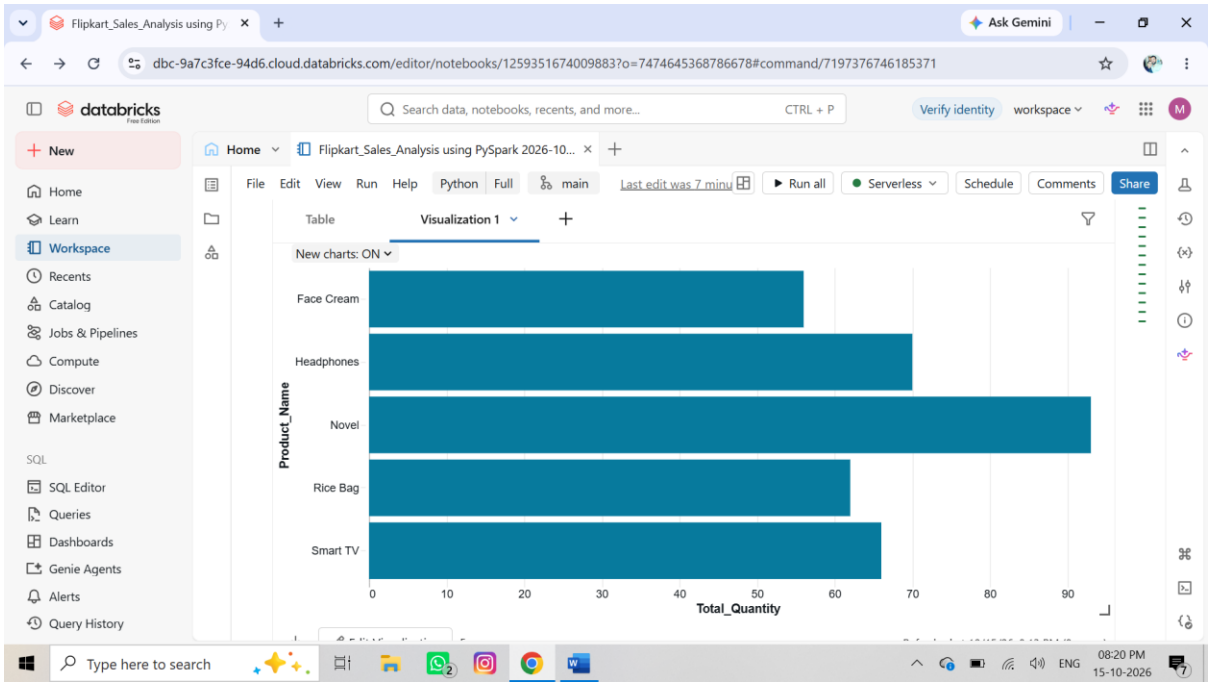


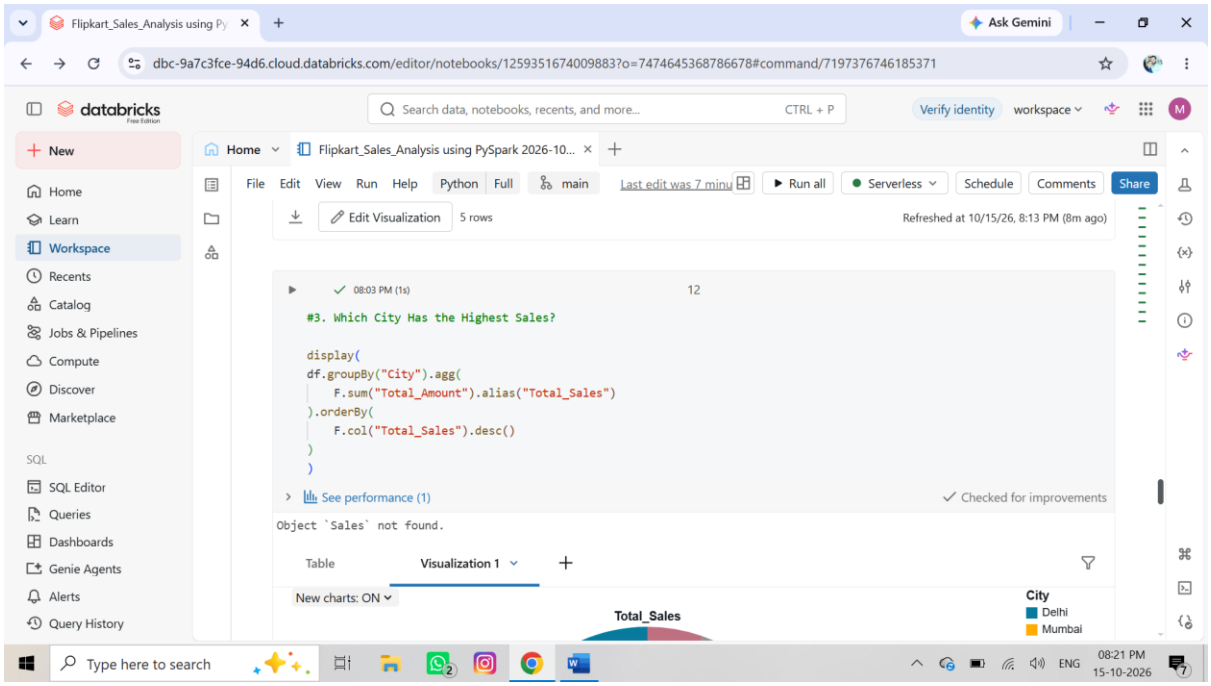
Filpkart Sales Analysis:

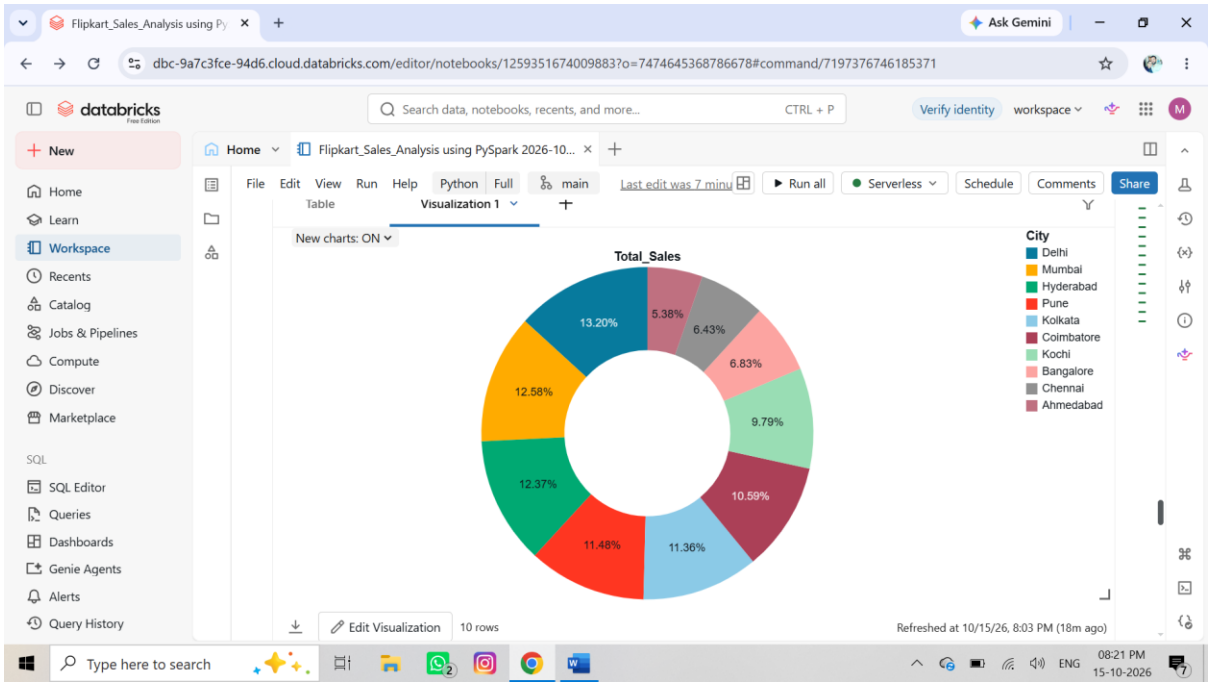


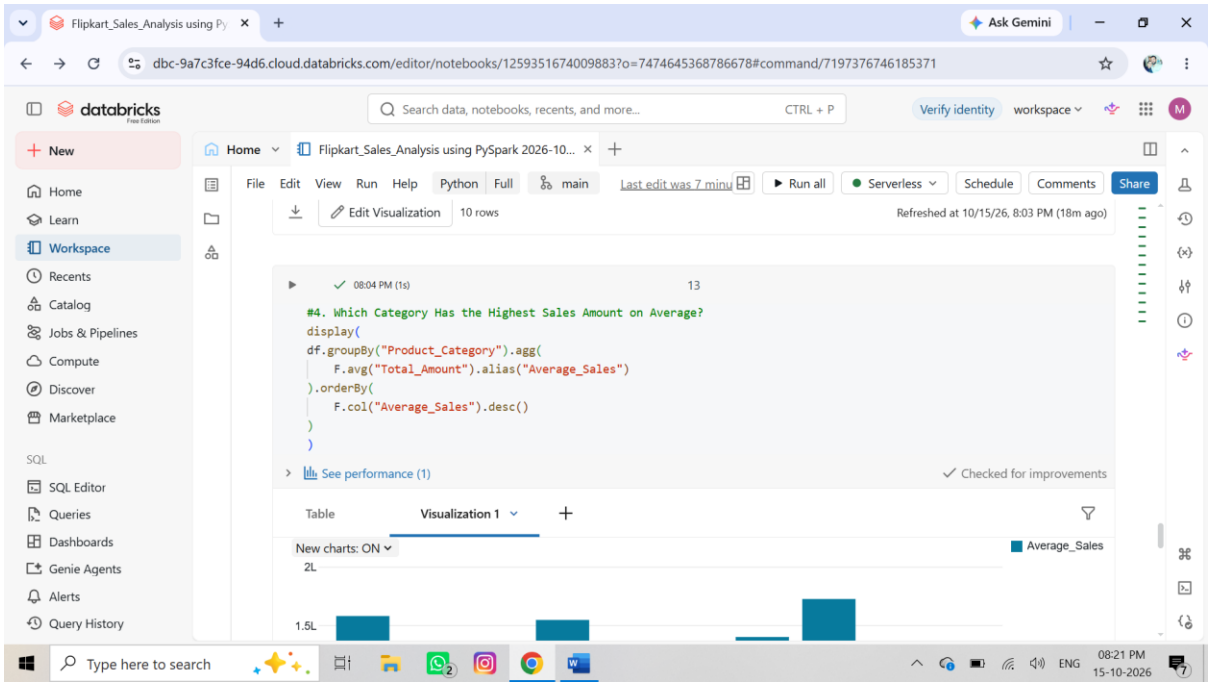

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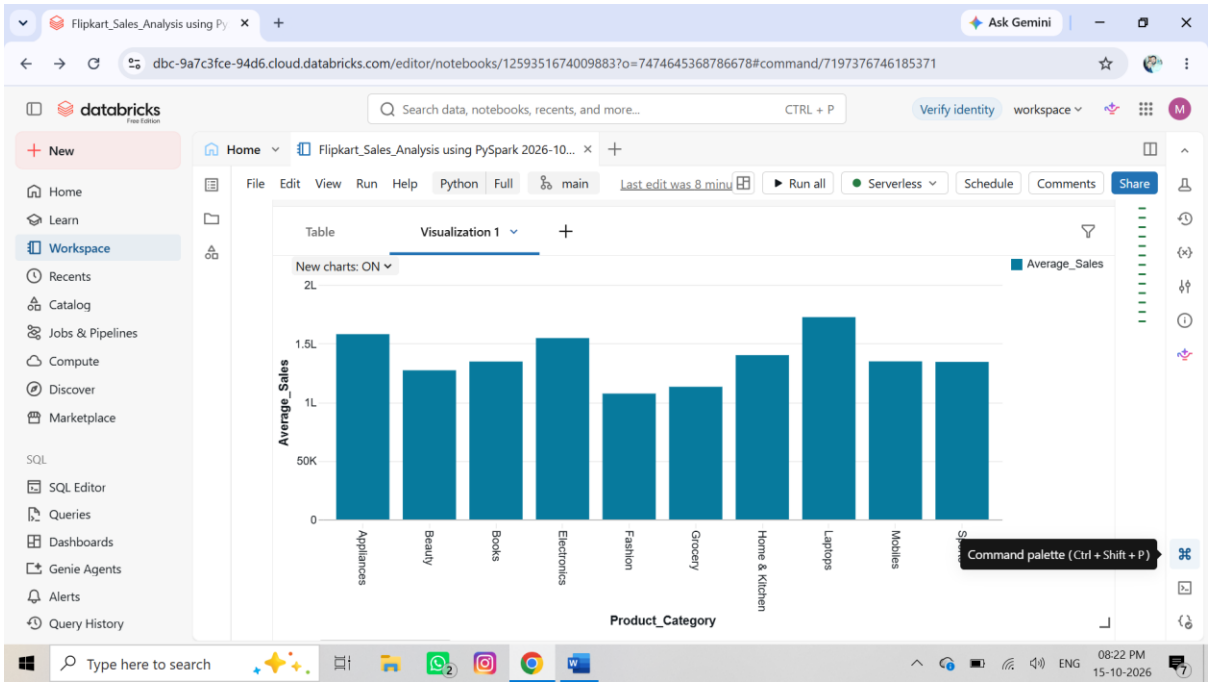

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#2. Which Product Sells the Most?
display(
df.groupBy("Product_Name").agg(
F.sum("Quantity").alias("Total_Quantity")
).orderBy(
F.col("Total_Quantity").desc()
).limit(5)
)
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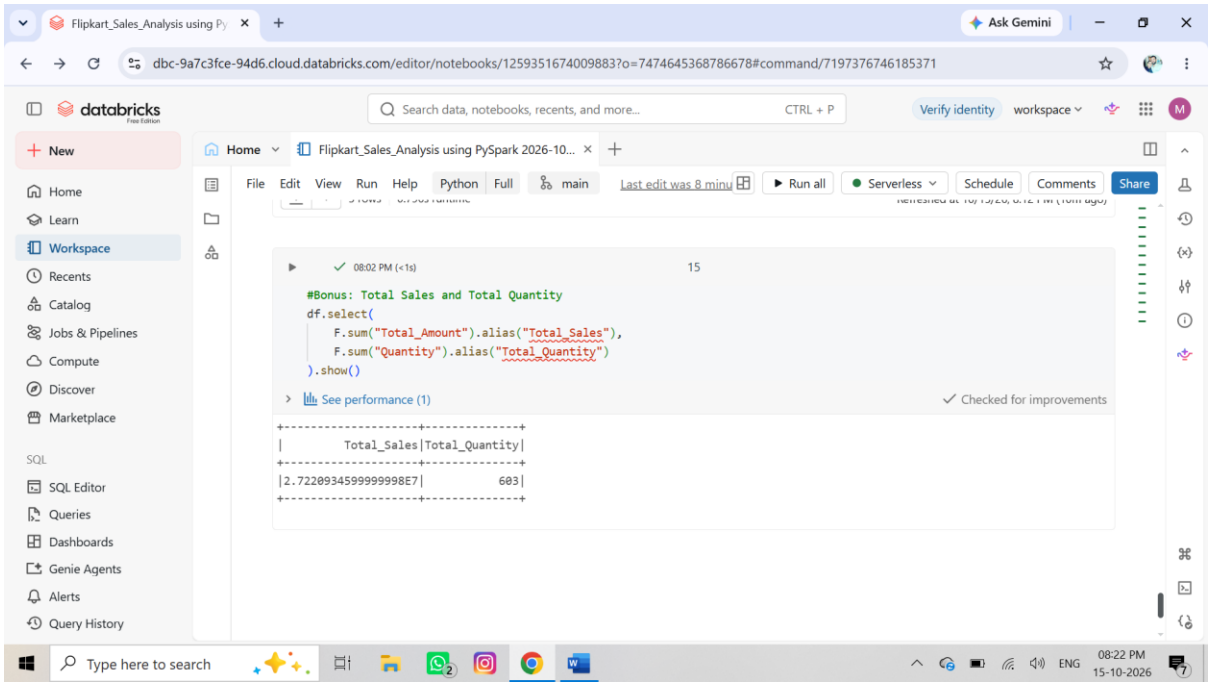

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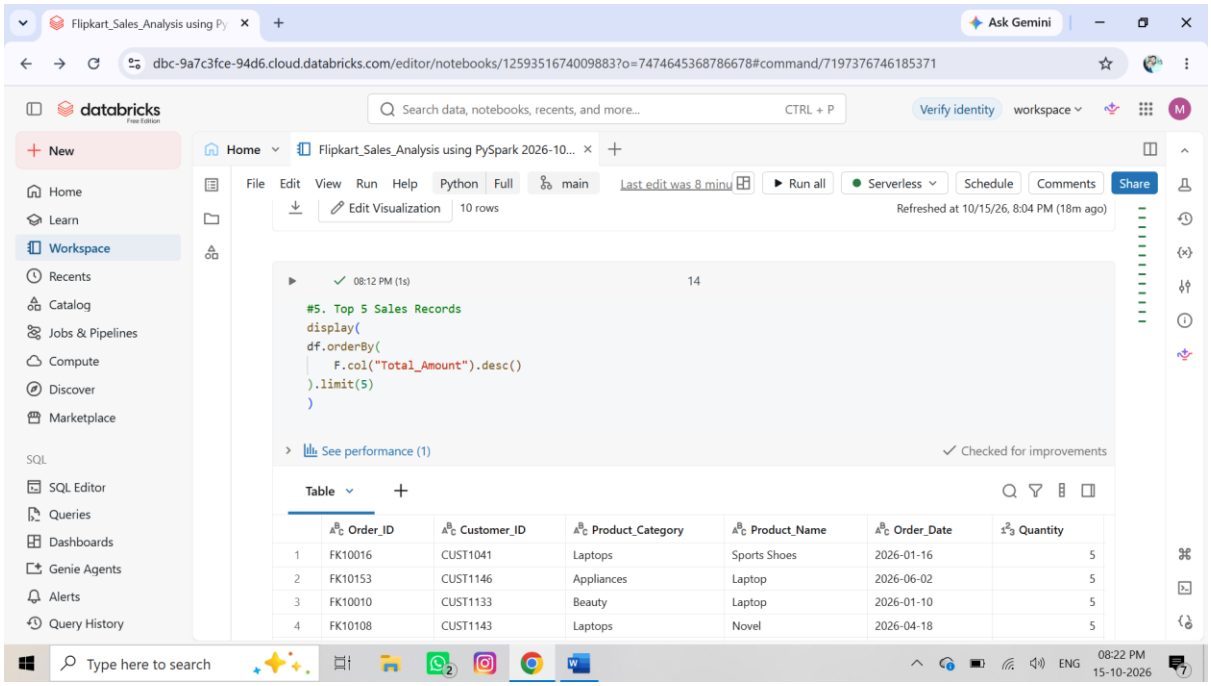

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#3. Which City Has the Highest Sales?
display(
df.groupBy("City").agg(
F.sum("Total_Amount").alias("Total_Sales")
).orderBy(
F.col("Total_Sales").desc()
)
)
See performance (1)
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Object `Sales` not found.
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5.38%
6.43%
6.83%
12.58%
9.79%
12.37%
10.59%
11.48%
11.36%
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Delhi
Mumbai
Hyderabad
Pune
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#4. Which Category Has the Highest Sales Amount on Average?
display(
df.groupBy("Product_Category").agg(
F.avg("Total_Amount").alias("Average_Sales")
).orderBy(
F.col("Average_Sales").desc()
)
)
See performance (1)
Checked for improvements
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#Bonus: Total Sales and Total Quantity
df.select(
F.sum("Total_Amount").alias("Total_Sales"),
F.sum("Quantity").alias("Total_Quantity")
).show()
See performance (1)
Checked for improvements
+--------------------+--------------+
| Total_Sales|Total_Quantity|
+--------------------+--------------+
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My Analysis

1. **Category-wise Sales Analysis:** Identify which product category generates the highest total sales.
2. **Product-wise Analysis:** Find the top-selling products based on quantity sold.
3. **City-wise Sales Analysis:** Identify the cities that generate the highest sales.
4. **Average Rating Analysis:** Compare average product ratings across categories, if rating data is available.
5. **Top 5 Sales Records:** Display the top five records based on sales value.
6. **Sales Performance:** Analyze total quantity sold and total sales to understand overall business performance.

These analyses help identify sales trends, customer purchasing patterns, and opportunities to improve business performance.

The Flipkart Sales Analysis project was developed using PySpark in Databricks to analyze e-commerce sales data.

The project covered data loading, data exploration, data cleaning, data transformation, business analysis, and visualization. PySpark functions were used to process and analyze the dataset efficiently.

The analysis helps identify high-performing product categories, popular products, cities with higher sales, and other useful business insights based on the available data.

The results can help businesses make better decisions regarding inventory management, product promotion, sales strategies, and customer satisfaction.

Overall, this project provided practical experience in using PySpark and Databricks for data analysis and demonstrated how data-driven insights can support business decision-making.

Conclusion:

This project demonstrates how PySpark can be used in Databricks to perform sales data analysis efficiently. The dataset was loaded, explored, cleaned, and transformed to prepare it for analysis. Business analysis helped identify sales trends, product performance, and opportunities to improve business decisions.Overall, this project provided practical experience in using Python, PySpark, and Databricks for data processing and sales analysis. The insights gained can help businesses improve sales performance and make data-driven decisions.